

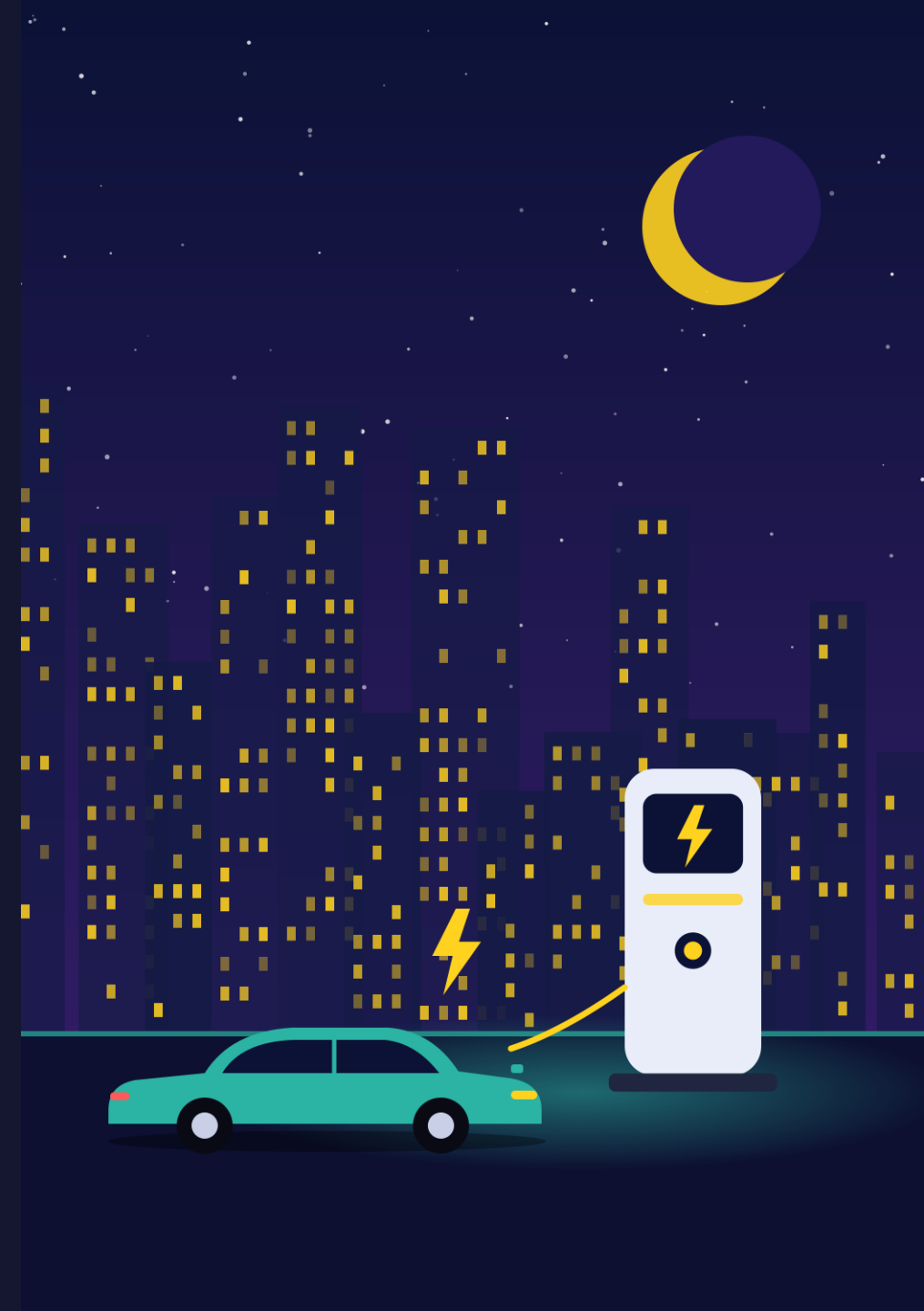


Powering the Future: A Data-Driven Expansion Strategy for Tata Power EV Networks

EV charging analytics: four dashboards on usage, stations, customers and ROI

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Cohort 136 | WS Cube Milestone Project



Introduction



- Tata Power is one of India's leading integrated power companies, spanning generation, transmission, distribution and renewables, and is scaling its EV charging network.
- Objective: turn operational, customer and financial data into clear insight on performance, utilisation and ROI, using data cleaning, visualisation and statistical analysis.
- Output: four Excel dashboards for senior management, strategy teams and operations managers, supporting decisions on expansion, pricing and infrastructure investment.

DATASET AT A GLANCE (AUG-NOV, 90 ACTIVE DAYS)

65

Stations

18

Districts

390

Plugs

1,14,000

Sessions



Problem Statement

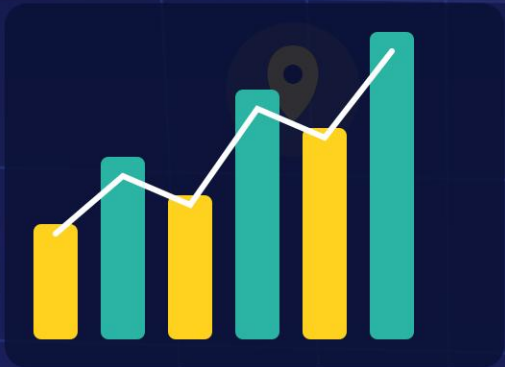
- Tata Power is expanding its EV charging network across districts, but operational and financial data is scattered, so tracking performance, profitability and utilisation is hard.
- That makes it difficult to spot underperforming stations and high-potential locations for expansion. An integrated Excel dashboard is needed to combine operations, customer usage and finance.

DECISIONS THE DASHBOARDS SUPPORT

Pricing

Performance
optimisation

Infrastructure
investment



Analysis Overview

1

Business and Usage Overview

Monitor charging demand, usage trend and overall network performance

2

Station Utilization and Operational Efficiency

Balance demand, improve plug utilisation and spot high-usage locations

3

Customer and Demand Behavior

Drive growth through different customer-segment engagement

4

Financial Performance and Expansion Strategy

Evaluate ROI and demand gaps for strategic expansion

1. Business and Usage Overview

BUSINESS PROBLEMS TO ADDRESS

- How is the overall charging demand evolving over time
- Is revenue aligned with energy consumption and pricing
- Are current stations able to handle the growing EV demand
- What is the relationship between number of customers and usage

KPIs TRACKED

Total charging sessions

1,14,000

Total energy consumed (kWh)

28,59,771.70

Total revenue

₹4,96,93,947.05

Active customers

8,000

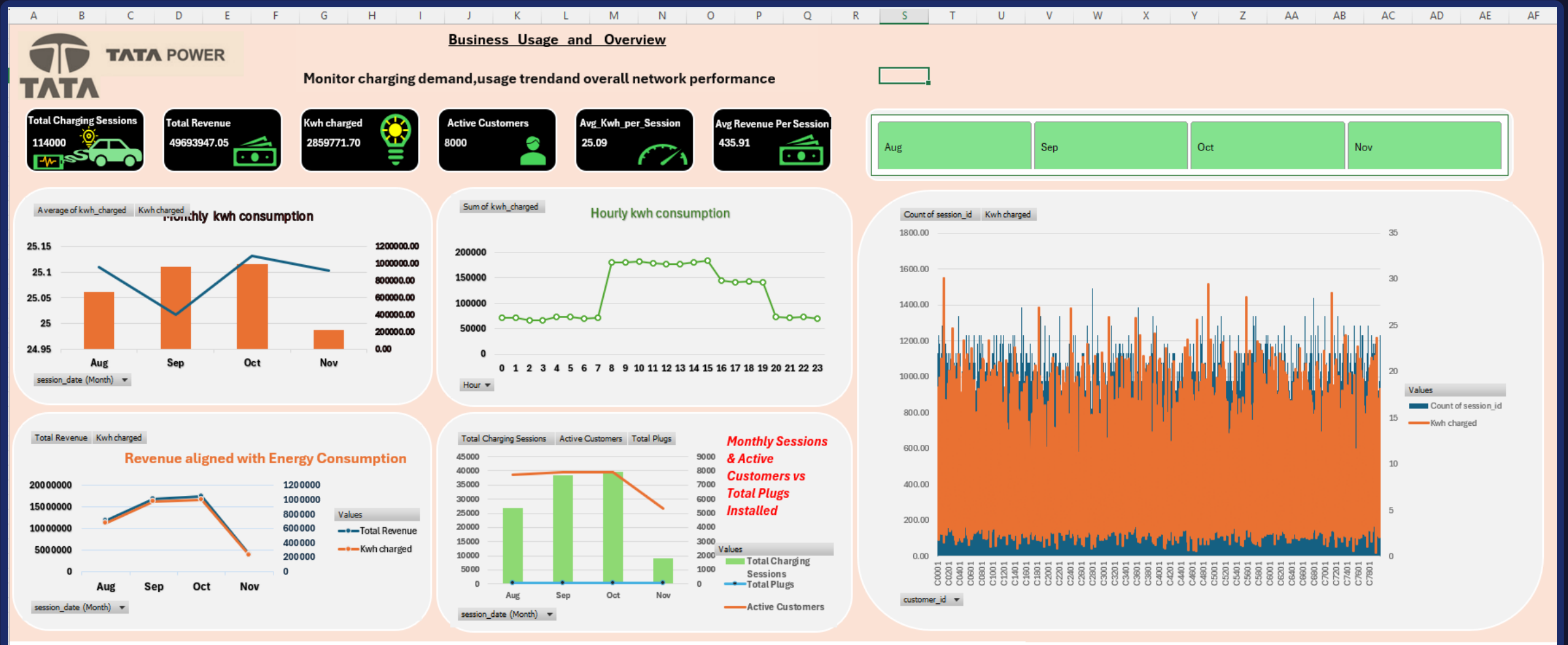
Avg revenue per session

₹435.91

Avg kWh per session

25.09

1. Business and Usage Overview



Monthly and hourly kWh:
demand trend and daily peak

Revenue vs energy: pricing
alignment

Sessions and active customers
vs plugs: capacity

Customer-level chart: usage per
customer

Business Insights

How is the overall charging demand evolving over time?

1

Demand grew ~50% from Aug to Oct and peaks from 8 AM to 4 PM

Energy rose from ~0.67M kWh (Aug) to ~1.0M (Oct); Nov (~0.22M) looks like a partial month. Average energy per session stays flat at ~25 kWh, so growth is more sessions. 8 AM–4 PM load is ~2.5x overnight.

Is revenue aligned with energy consumption and pricing?

2

Yes: revenue tracks energy almost one-to-one

Revenue and kWh rise and fall together every month, which implies a flat effective price of about ₹17.4 per kWh (₹435.91 per session ÷ 25.09 kWh). No month shows revenue lagging volume.

Are current stations able to handle the growing EV demand?

3

Not for long: sessions grew ~47% while plugs stayed at 390

Monthly sessions climbed from ~27k (Aug) to ~40k (Oct) on an unchanged plug base, with active customers flat at ~7.7k–7.9k. Growth is coming from more frequent charging, so capacity is the next constraint.

What is the relationship between number of customers and usage?

4

Usage is broad-based and driven by charging frequency

8,000 customers average 14 sessions and ~357 kWh each; the customer-level chart shows no single dominant account, only a few heavy-user spikes. Sessions per customer rose from ~3.5 to ~5 a month (Aug to Oct).

2. Station Utilization and Operational Efficiency

BUSINESS PROBLEMS TO ADDRESS

- Which stations are overutilized and may require capacity expansion
- Which stations are underutilized and need operational improvement
- Are available plugs being efficiently used across locations
- What is the load distribution across districts and stations

KPIs TRACKED

Sessions per station

1,753.85

Total plugs

390

Avg sessions per plug per day

3.25

Number of stations

65

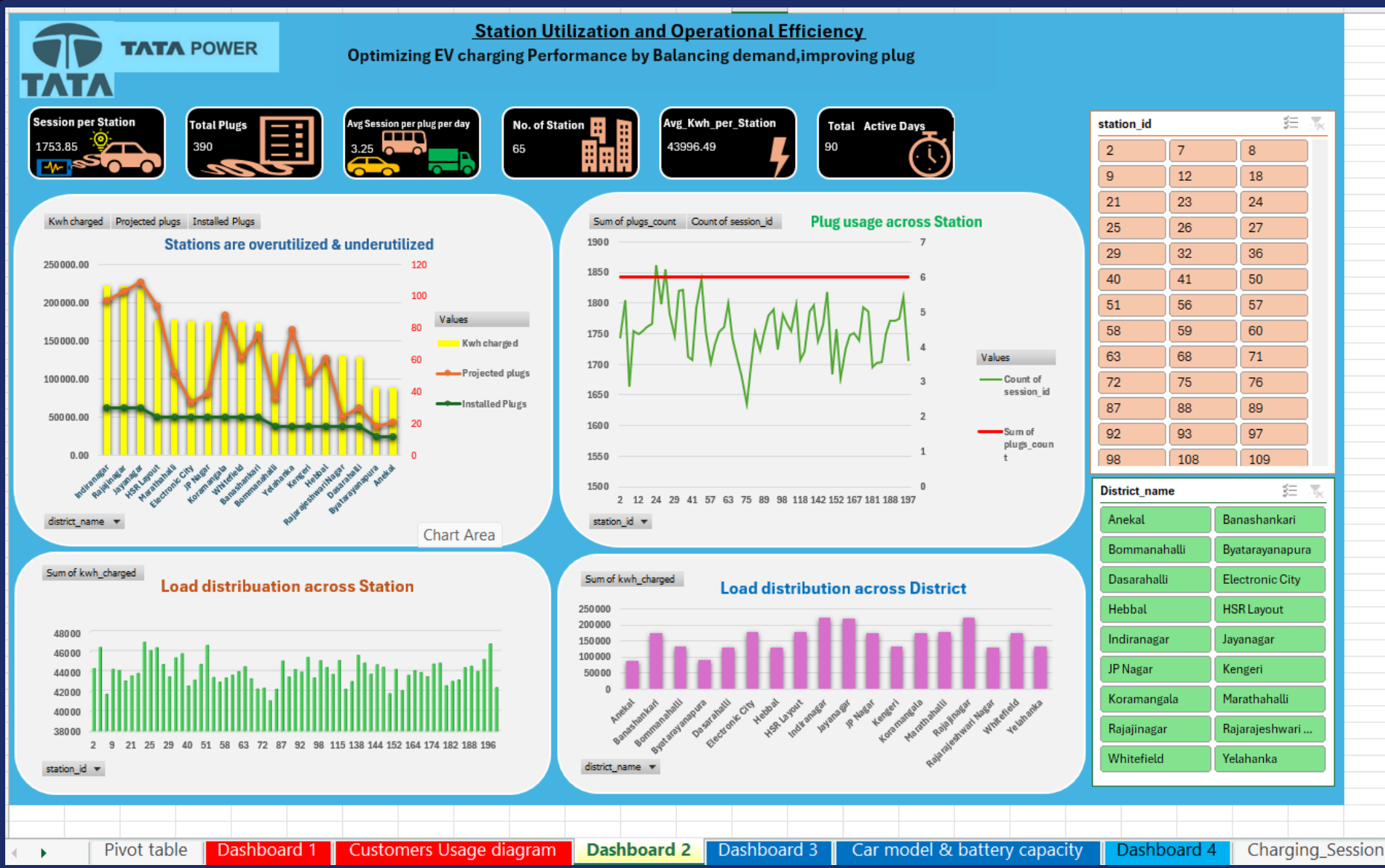
Avg kWh per station

43,996.49

Total active days

90

2. Station Utilization and Operational Efficiency



HOW TO READ IT

- Top left: kWh by district against installed and projected plugs
- Top right: sessions per station against the 6-plug line
- Bottom: load by station and by district
- Slicers: station and district

Station Insights

1

Which stations are overutilized and may require capacity expansion?

Indiranagar, Rajajinagar and Jayanagar need capacity first

They carry the highest load (~220k kWh each) on just 30 installed plugs, against ~100+ projected plugs. At station level the busiest sit around IDs 24–26 (~1,860 sessions, ~6% above average).

2

Which stations are underutilized and need operational improvement?

Anekal and Byatarayanapura are the lowest-volume districts

~85–90k kWh each on 12 plugs (2 stations). Their per-station load is in line with the network, so the gap is footprint, not station performance. The weakest single stations (~41k kWh, near ID 72) are only ~7% under average.

3

Are available plugs being efficiently used across locations?

Plug use is uniform: 6 plugs per station, 3.25 sessions per plug per day

Sessions per station sit in a narrow ~1,630–1,860 band (average 1,754, about $\pm 7\%$), with 292 sessions per plug across 90 active days. Uniform use means expansion should follow demand, not fix weak stations.

4

What is the load distribution across districts and stations?

Balanced by station, uneven by district

Most stations deliver 42–46k kWh (average 43,996), although the chart axis starts at 38k and exaggerates gaps. District totals range from ~85k to ~222k kWh because districts host 2 to 5 stations each (65 in total).

3. Customer and Demand Behavior

BUSINESS PROBLEMS TO ADDRESS

- What type of customers contribute most to usage and revenue
- How does income tier influence charging behavior
- What are the usage patterns across different EV models and battery capacities
- Are there segments with high potential but low engagement

KPIs TRACKED

Unique customers

8,000

Sessions per customer

14

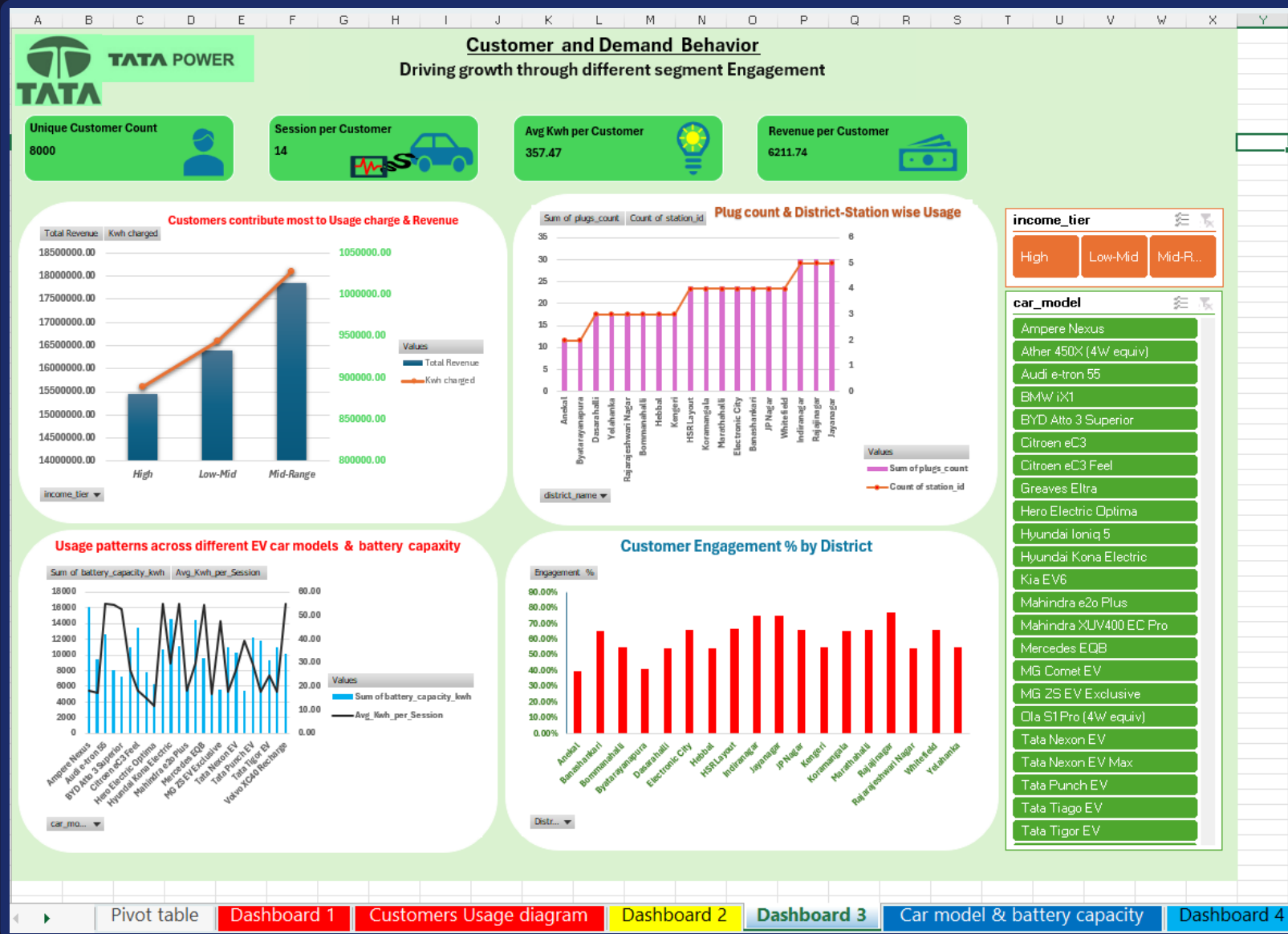
Avg kWh per customer

357.47

Revenue per customer

₹6,211.74

3. Customer and Demand Behavior



HOW TO READ IT

- Top left: revenue and kWh by income tier
- Top right: plugs and stations per district
- Bottom left: usage by EV model and battery capacity
- Bottom right: customer engagement % by district
- Slicers: income tier and car model

Customer Insights

1

What type of customers contribute most to usage and revenue?

Mid-Range income tier leads, with ~36% of revenue and energy

Mid-Range ≈ ₹1.78 Cr and ~1.02M kWh; Low-Mid ≈ ₹1.64 Cr and ~0.95M kWh; High ≈ ₹1.55 Cr and ~0.89M kWh. The chart axes start above zero, so the gaps look bigger than the ~15% real spread.

2

How does income tier influence charging behavior?

Influence is modest: every tier holds 31–36% of usage

Usage rises from High to Low-Mid to Mid-Range, but no tier dominates. The High tier charges least despite the greatest purchasing power, possibly because of private home charging, a hypothesis to validate.

3

What are the usage patterns across different EV models and battery capacities?

Large-battery cars draw up to ~55 kWh a session, small EVs ~12–18

Audi e-tron 55, Mercedes EQB and Volvo XC40 Recharge peak near 55 kWh per session, while Hero Electric Optima (~12) and Ampere Nexus (~18) sit at the low end. Model mix therefore shapes session length and plug time.

4

Are there segments with high potential but low engagement?

Anekal (~39%) and Byatarayanapura (~41%) are the least engaged

Engagement peaks in Rajajinagar (~77%), Indiranagar and Jayanagar (~75%); six districts sit near 54–55%. The two lowest also have the smallest footprint (2 stations, 12 plugs), and with the High income tier they are the main campaign targets.

4. Financial Performance and Expansion Strategy

BUSINESS PROBLEMS TO ADDRESS

- Which districts and projects deliver the highest ROI
- What is the payback period for different charging setups
- How do costs such as CAPEX, land lease, and infrastructure impact profitability
- Which locations should be prioritized for new station deployment based on demand and ROI
- What is the gap between existing infrastructure and projected EV demand

KPIs TRACKED

Total CAPEX

₹13,921.62

Lakh

Annual revenue

₹4,628.64

Lakh

Annual EBITDA

₹3,238.53

Lakh

5-year ROI

41.43%

Avg payback (without subsidy)

4.61

years

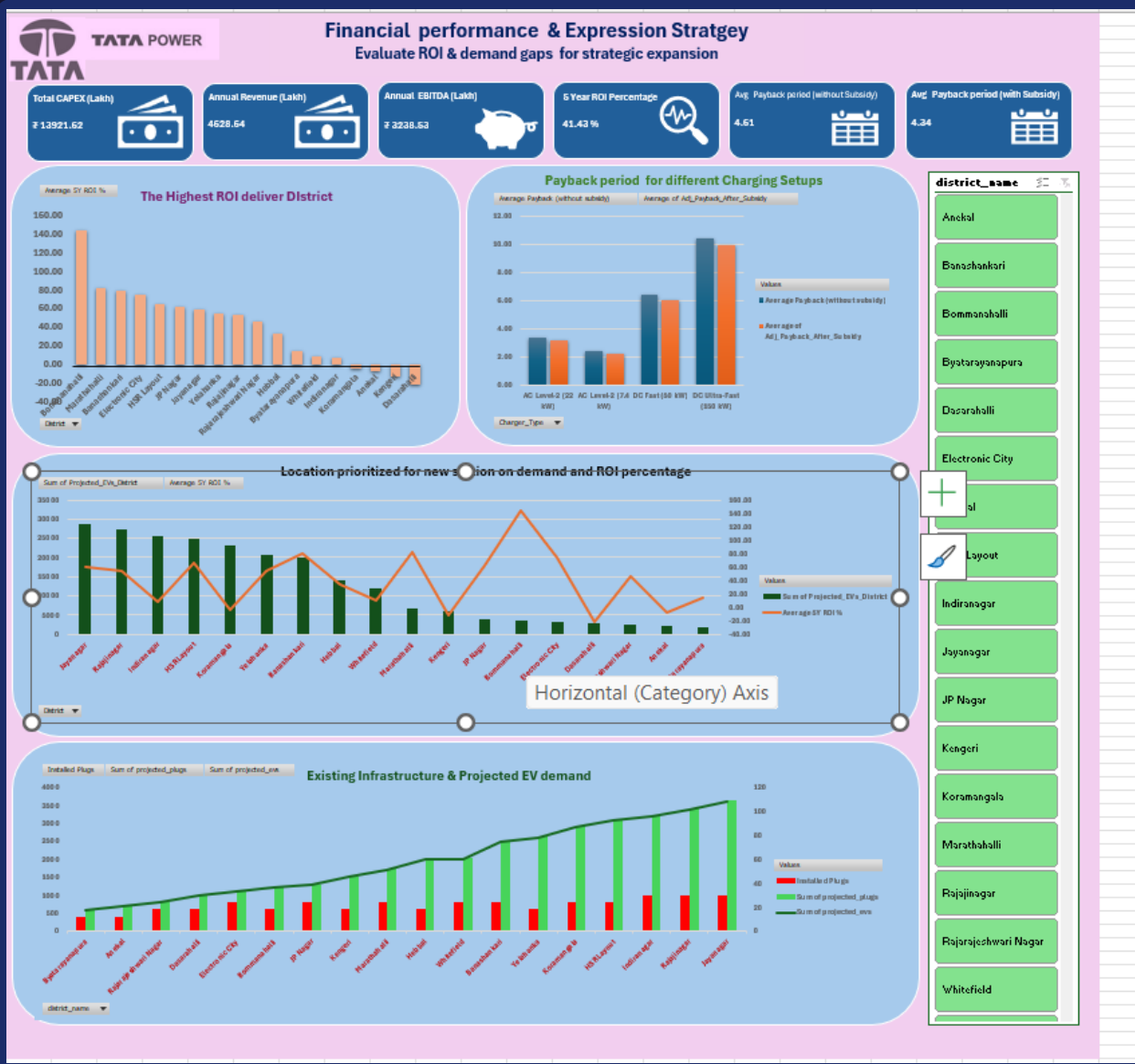
Avg payback (with subsidy)

4.34

years

Derived: EBITDA margin $\approx 70\%$ ($3,238.53 \div 4,628.64$); subsidy shortens average payback by ~ 0.27 years; CAPEX $\approx$ ₹142 Lakh per project (PDF analysis).

4. Financial Performance and Expansion Strategy



HOW TO READ IT

- Top left: 5-year ROI % by district
- Top right: payback years by charger type, with and without subsidy
- Middle: projected EVs against ROI to prioritise locations
- Bottom: installed plugs vs projected plugs and EVs
- Slicer: district

Financial Insights

1

Which districts and projects deliver the highest ROI?

Bommanahalli leads at ~145% 5-year ROI; four districts are negative

Marathahalli (~82%), Banashankari (~80%) and Electronic City (~75%) follow, and 14 of 18 districts are ROI-positive. Koramangala, Anekal, Kengeri and Dasarahalli fall below zero (about -3% to -22%).

2

What is the payback period for different charging setups?

AC 7.4 kW pays back in ~2.3 years, DC Ultra-Fast in ~10

AC 22 kW ≈ 3.3 yrs, DC Fast 50 kW ≈ 6.4 yrs, DC Ultra-Fast 150 kW ≈ 10.4 yrs; subsidy trims 0.2–0.5 yrs. The network average is 4.61 years, or 4.34 with subsidy.

3

How do costs such as CAPEX, land lease, and infrastructure impact profitability?

Capital intensity drives payback; operating margin is strong (~70%)

EBITDA of ₹3,238.53 Lakh on ₹4,628.64 Lakh revenue sits against ₹13,921.62 Lakh CAPEX. Ultra-fast chargers take ~4.5x longer to repay than 7.4 kW AC. Land lease is not broken out in the dashboard.

4

What is the gap between existing infrastructure and projected EV demand?

Installed plugs stay flat while projected demand climbs steeply

Installed plugs stay at ~12–30 per district, while projected plugs reach ~97–108 and projected EVs ~3,300–3,500 in Jayanagar, Rajajinagar and Indiranagar. Network-wide, that is ~33 projected EVs per projected plug.

Priority Tier Classification

Tier 1 Expand now

High demand, ROI $\geq$ ~50%

Jayanagar · 28.5k EVs · 59%

Rajajinagar · 27k · 53%

HSR Layout · 24.5k · 65%

Yelahanka · 20.5k · 55%

Banashankari · 20k · 80%

Tier 2 Fix, then expand

High demand, ROI $<$ ~35%

Indiranagar · 25.5k · 7%

Koramangala · 23k · -3%

Hebbal · 14k · 33%

Whitefield · 12k · 9%

Tier 3 Targeted, phased adds

*ROI $\geq$ ~45%, demand $<$ ~7k
EVs*

Bommanahalli · 3.6k · 145%

Marathahalli · 6.7k · 82%

Electronic City · 3.3k · 75%

JP Nagar · 3.9k · 62%

Rajarajeshwari Nagar · 2.5k
· 46%

Tier 4 Hold and monitor

Low demand, ROI $<$ ~15%

Kengeri · 6k · -13%

Dasarahalli · 2.9k · -22%

Anekal · 2.1k · -8%

Byatarayanapura · 1.7k ·
14%

Each line: district · projected EVs · 5-year ROI. Derived from the "Location prioritized for new station" and "Highest ROI" charts; values are read from the charts and approximate.

Conclusion

1,14,000

charging sessions

₹4.97 Cr

session revenue

41.43%

5-year ROI

- **Demand is growing and monetising well.** Sessions rose ~47% from Aug to Oct, revenue tracks energy at ~₹17.4 per kWh, and the customer base is broad (8,000, no dominant account).
- **Capacity is the constraint where demand is highest.** Indiranagar, Rajajinagar and Jayanagar run 30 plugs against ~100+ projected; per-station load is even, so the answer is more plugs, not fixing stations.
- **Returns are uneven.** ROI spans ~145% (Bommanahalli) to -22% (Dasarahalli), and payback ranges from ~2.3 years (AC 7.4 kW) to ~10 years (DC Ultra-Fast).
- **Focus investment by tier.** Expand Tier 1 now, fix economics in high-demand Tier 2, phase Tier 3 adds and hold Tier 4, using AC and DC Fast chargers for daily use.

Plan for the Coming Year

Infrastructure

- Add plugs first in Jayanagar, Rajajinagar and Indiranagar, where installed plugs sit far below projected demand.
- Pre-install extra conduit and electrical capacity at new sites so plugs can be added later without site overhauls.
- Move to usage-based maintenance for the busiest stations (around IDs 24–26, ~1,860 sessions each).

Financial expansion

- Hold new CAPEX in negative-ROI districts (Anekal, Dasarahalli, Kengeri) and diagnose Koramangala and Indiranagar first.
- Reallocate capital to Tier 1 districts and to high-ROI Bommanahalli in phased steps.
- Favour AC and DC Fast chargers for daily use; keep DC Ultra-Fast for selective high-traffic sites.

Customers and demand

- Introduce time-of-use pricing to shift load from the 8 AM–4 PM peak to off-peak hours.
- Run loyalty and engagement campaigns for the High income tier and for Anekal and Byatarayanapura.
- Target Mid-Range neighbourhoods for new sites, as they deliver the most energy and revenue.



THANK YOU

Questions are welcome

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